25901105

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FOURTH SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION, MARCH 2026

Discipline Specific Elective course- MG4DSEBTS200

ENZYMOLOGY

(2024 Admission Onwards)

Time: 2 Hours **Max. Marks: 70**

Part A

Multiple Choice Questions

Answer all questions.

Each question carries 1 mark

1. (b) Isomerases [U CO1]
2. (b)To prevent enzyme degradation by proteases.[U CO3]
3. (b) Isolate the enzyme from other cellular components while retaining activity [U CO3]
4. (c) Heart Disease [K CO4]
5. (b) They require proteolytic cleavage for action [U CO1]
6. (b) Covalent Regulation [K CO2]
7. (b) Increasing the activation energy for substrate conversion [A CO2]
8. (c) Product is difficult to detect directly [U CO3]
9. (c) Pancreas [U CO4]
10. (b) To provide uniform negative charge and denature proteins[U CO3]

Part B

Short Answer Questions

Answer any ten questions.

Each question carries 3 marks.

11. Definition- Biological catalysts enhancing rate of reaction- 1 mark
    Any two general properties from nature (protein),lowering activation energy, factors affecting activity, transition state - 1 mark each
    (U, CO1)

12. Definition of serine proteases with presence of catalytic triad (serine, histidine, aspartate- 1 mark
    Examples in digestion (trypsin,chymotrypsin,elastase-1 mark)
    Cleavage of peptide bonds, zymogen activation in protein digestion-1 mark
    [K,CO2]

13. Any 3 biophysical techniques with a brief description (1 mark each)
    From the following- X ray crystallography, Centrifugation, Any spectroscopic technique,Electrophoresis etc
    [U,CO3]

14. Diagnostic marker for pancreatic Disorders (1 mark)
    Monitoring of gastrointestinal disorders (1 mark)
    Analysis of salivary gland disorders and diabetes (1mark)
    [K, CO4]

15. Differentiating isoenzymes involves separating and identifying variants of the same enzyme on the basis of charge,size or other physicochemical properties (1 mark)
    Any 2 types of sample preparation, separation and visualizing or identifying methods for isoenzymes preferably electrophoresis (2 marks)
    [U,CO4]

16. X ray crystallography determines and elucidates 3D structure of enzymes & proteins (1 mark)
    Principle- Braggs law (0.5 mark)
    Process-including crystallization (0.5 mark)
    Applications- finding active site, structure validation etc (1 mark)
    [U,CO3]

17. FAD- Flavin Adenine Dinucleotide derived from Vitamin Riboflavin (1 mark)
    Role in oxidation reduction reactions acting as coenzyme (1.5 marks)
    Examples Succinate to Fumarate in TCA cycle (0.5 mark)
    [K,CO2]

18. Ultra centrifugation uses high speed rotation of $1X\ 10^6$ G for separation of enzymes based on sedimentation rates determined by size , shape and density. (1 mark)
    Principle – (0.5 mark)
    Types- (0.5 mark)
    Applications- determining molecular weight,purity and binding interactions of enzymes- (1 mark)
    [U,CO3]

19. Detergent industry- (1 mark)
    Food processing industry – (1 mark)
    Biodiesel production- (1 mark)
    [K,CO4]

20. Competitive inhibition- Substrate competes with inhibitor for active sites on enzyme (1 mark)
    Substrate should be structural analogue of inhibitor with a suitable example. (1 mark)
    Km value of substrate is lesser for enzyme in presence of enzyme, V max does not change, LB plot (1 mark)
    [K,CO2]

21. Lowering activation energy by molecular interactions with substrate enabling reactions at standardized conditions (1 mark)

Active site binding (1 mark)
Acid base catalysis- microenvironment effects including mechanisms (1 mark)
[U,CO1]

22. Enzyme kinetics helps to study how enzymes function and regulate metabolic pathways (1 mark)
Determine values of Km,Vmax to understand binding state mechanisms (1 mark)
Support drug development and diagnostics (1 mark)
[AN,CO2]

Part C

Short Essay Questions

Answer any 6 questions.

Each question carries 5 marks

23. Fischers Lock & Key- Diagram (2 marks)
Hypothesis and its limitation including the case of inability to explain the action of different enzymes on the same substrate and competitive inhibition (2 marks)
Historical importance and why Koshlands model was preferred (1 mark)
[U,CO1]

24. Cooperativity and feedback inhibition are key regulatory mechanisms in enzymes that enable control of metabolic pathways. (1 mark)
Cooperativity- Binding of modulators on the regulatory subunit alter the activity of enzyme in the active site, Either positively or negatively, One example can be given,Sigmoid kinetics (2 marks)
End product of the pathway allosterically inhibits the activity of the key enzyme, non competitive inhibition alters shape of the active site, provides pathway specific control, one suitable example. (2 marks)
[K,CO2]

25. TLC separates compounds based on differential adsorption beteween a stationary phase (silica gel on a plate) and a mobile solvent phase, visualized as spots that can be differentiated on the basis of Rf values. (1 mark)
Process and applications (3 marks)
Advantages over paper chromatography- Faster separation, higher sensitivity (1 mark)
[U,CO3]

26. Active site – Definition- must include that it’s a site formed by contributing amino acids uniquely joining from the 3D structure of protein which align their groups based on the structure of substrate to form a cleft/crevice/groove. (1 mark)
Catalytic activity first step is binding to substrate.(1 mark)
Methods of mapping any two with description from Affinity Labelling, Chemical Labelling, Site Directed mutagenesis or Bioinformatics used (3 marks)
[U,CO1]

27. Active site – Definition- must include that it's a site formed by contributing amino acids uniquely joining from the 3D structure of protein which align their groups based on the structure of substrate to form a cleft/crevice/groove. (1 mark)
Catalytic activity first step is binding to substrate.(1 mark)
Methods of mapping any two with description from Affinity Labelling, Chemical Labelling, Site Directed mutagenesis or Bioinformatics used (3 marks)
[K,CO2]

28. Enzyme extraction from cells involve disrupting mechanisms of cellular structures to release intracellular enzymes preserving their activity following purification methods.
(1 mark each for the methods with a brief description)
Mechanical Disruption
Osmotic Shock
Enzymatic Lysis
Freeze Thaw Cycles
Chemical Methods
[U,CO3]

29. Fermentation plays a crucial role in the food industry by producing preserved, flavorful products through microbial action on sugars and other substrates.

- **Yogurt**: Lactic acid bacteria (Lactobacillus bulgaricus, Streptococcus thermophilus) ferment lactose in milk, creating tangy flavor, texture, and probiotics
- **Cheese**: Bacteria, yeast, and molds ferment milk sugars and proteins over weeks/months, developing complex flavors and textures (e.g., cheddar, gouda).
- **Beer**: Yeast ferments barley malt sugars into ethanol and CO_2 via alcoholic fermentation, producing carbonation and alcohol content.
- **Wine**: Yeast converts grape sugars to ethanol and CO_2, with malolactic fermentation by bacteria softening acidity for red/white varieties.
- **Sauerkraut/Kimchi**: Lactic acid fermentation of cabbage by Leuconostoc and Lactobacillus creates sour taste, preservation, and gut-healthy probiotics.

(1 mark each for each product) [K,CO4]

30. Holoenzymes represent the complete, catalytically active form of enzymes that require non-protein components for function.

    - Complete enzyme complex consisting of apoenzyme (protein part) plus cofactor (inorganic ion or organic coenzyme like NAD+), enabling full catalytic activity.
    - Apoenzyme alone is inactive; cofactor binding induces proper conformation and stabilizes transition states for substrate catalysis.
    - Examples include RNA polymerase holoenzyme (with sigma factor) for transcription initiation and DNA polymerase III for replication.
    - Cofactors often derived from vitamins (e.g., FAD from riboflavin, CoA from pantothenic acid) act as transient carriers of electrons, acyl groups, or hydrides.
    - Essential in metabolic pathways; dissociation into apoenzyme yields inactive form, providing regulatory control. [.U,CO1]